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ORIGINAL ARTICLE

Long-term kidney transplant graft survival—Making progress when most needed

Emilio D. Poggio[1,2] | **Joshua J. Augustine**[1,2] | **Susana Arrigain**[3,4] | **Daniel C. Brennan**[5] | **Jesse D. Schold**[2,3,4]

[1]Department of Nephrology and Hypertension, Glickman Urological and Kidney Institute, Cleveland Clinic, Cleveland, Ohio

[2]Cleveland Clinic Lerner College of Medicine, Case Western Reserve University, Cleveland, Ohio

[3]Department of Quantitative Health Sciences, Cleveland Clinic, Cleveland, Ohio

[4]Center for Populations Health Research, Lerner Research Institute, Cleveland Clinic, Cleveland, Ohio

[5]Division of Nephrology, Johns Hopkins University, Baltimore, Maryland

Correspondence
Emilio D. Poggio, Department of Nephrology and Hypertension, Glickman Urological and Kidney Institute, Cleveland Clinic, Cleveland, OH, USA.
Email: poggioe@ccf.org

Current short-term kidney post–transplant survival rates are excellent, but longer-term outcomes have historically been unchanged. This study used data from the national Scientific Registry of Transplant Recipients (SRTR) and evaluated 1-year and 5-year graft survival and half-lives for kidney transplant recipients in the US. All adult (≥18 years) solitary kidney transplants (*n* = 331,216) from 1995 to 2017 were included in the analysis. Mean age was 49.4 years (SD +/-13.7), 60% male, and 25% Black. The overall (deceased and living donor) adjusted hazard of graft failure steadily decreased from 0.89 (95%CI: 0.88, 0.91) in era 2000–2004 to 0.46 (95%CI: 0.45, 0.47) for era 2014–2017 (1995–1999 as reference). Improvements in adjusted hazards of graft failure were more favorable for Blacks, diabetics and older recipients. Median survival for deceased donor transplants increased from 8.2 years in era 1995–1999 to an estimated 11.7 years in the most recent era. Living kidney donor transplant median survival increased from 12.1 years in 1995–1999 to an estimated 19.2 years for transplants in 2014–2017. In conclusion, these data show continuous improvement in long-term outcomes with more notable improvement among higher-risk subgroups, suggesting a narrowing in the gap for those disadvantaged after transplantation.

KEYWORDS
clinical research / practice, epidemiology, health services and outcomes research, kidney transplantation / nephrology, registry / registry analysis, risk assessment / risk stratification

1 | INTRODUCTION

Kidney disease is a common chronic medical condition with significant negative implications to the healthcare community. In 2019, the federal government signed the Executive Order "Advancing American Kidney Health" in which one of the objectives is to facilitate and promote kidney transplantation as the optimal modality for renal replacement therapy.[1,2] Also in 2019, UNOS/OPTN reported a record number of kidney transplants performed in the United States. This number was projected to be even higher for 2020 until the COVID-19 pandemic attenuated transplant rates in the United States and globally.[3] The increase was a consequence of a record number of both living and deceased donor transplants.[4] Kidney transplantation is the best available therapy for patients suffering from advanced chronic kidney disease or end stage kidney disease (ESKD),[5,6] because it does not only provide better quality of life compared to dialysis but also prolongs patient survival.[7,8] In this regard, kidney transplant recipients on average experience one-year graft and patient survival rates exceeding 95%.[9-11] These excellent short-term outcomes are difficult to improve upon, but paradoxically have not

Abbreviations: BMI, body mass index; ESKD, end stage kidney disease; KDRI, Kidney Donor Risk Index; OPTN, Organ Procurement and Transplantation Network; SRTR, Scientific Registry of Transplant Recipients; UNOS, United Network for Organ Sharing.

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correlated with improvement in longer-term survival, according to previous reports.[10-12]

Advances in the care of transplant recipients such as improvements in organ procurement and surgical techniques, development of immunosuppression regimens targeting early rejection, and prophylactic antibiotic therapies among others, have led to excellent one-year post–transplant survival rates. However, it is unclear whether these interventions along with advances in the care of patients with cardiovascular disease, diabetes, malignancies, and infectious diseases, have contributed to more recent improvements in longer-term outcomes. Therefore, the objectives of this study are to evaluate changes in short- and long-term kidney transplant outcomes over time in the United States, and to study whether these outcomes have evolved differently by patient demographics and clinical characteristics.

2 | METHODS

The data reported here have been supplied by the Hennepin Healthcare Research Institute (HHRI) as the contractor for the Scientific Registry of Transplant Recipients (SRTR). The interpretation and reporting of these data are the responsibility of the author(s) and in no way should be seen as an official policy of or interpretation by the SRTR or the U.S. Government. This study used the SRTR database as updated by March 2020. The SRTR data system includes data on all donor, waitlisted candidates, and transplant recipients in the US, submitted by the members of the Organ Procurement and Transplantation Network (OPTN), and has been described elsewhere.[13] The Health Resources and Services Administration (HRSA), U.S. Department of Health and Human Services provides oversight to the activities of the OPTN and SRTR contractors. This study was exempted from approval by the Cleveland Clinic Institutional Review Board (IRB# 09–648).

We included all adult (age >18) single organ kidney transplant recipients from 1995 to 2017 with last follow-up as of March 2020. Patients may have received more than one transplant (re-transplant), and therefore may have been represented repeatedly during the study period. For the purpose of the analyses, we grouped the patients into eras of transplantation as follows: from year 1995 to 1999, 2000 to 2004, 2005 to 2009, 2010 to 2013 and 2014 to 2017. We summarized donor and recipient characteristics by era of transplant. We evaluated trends of donor and recipient characteristics (outlined below) using linear regression for continuous variables, Cochran-Armitage Trend Test for binary variables, and Cochran-Mantel-Haenszel Statistics for categorical variables with more than two levels.

Our primary outcome was time to graft failure, a composite of the earlier of either graft loss or death. Secondary outcomes were graft failure censored at death, death censored at graft failure, graft failure with death as a competing risk, and death with graft failure as a competing risk. Administrative censoring of the data was on March 1, 2020. We created Kaplan-Meier graft survival plots by era of transplant. We tabulated estimates at different time points as well as the median graft survival, and graphed the survival curves. For more recent eras, where median graft survival was not observed, we estimated median graft survival times using linear regression of time versus observed Kaplan-Meier estimates. We estimated 95% confidence intervals around the predicted half-lives of deceased and live donor kidney transplant recipients separately by creating 1,000 bootstrap datasets with replacement, and conducting the regression analysis on each of them. Then we summarized the 2.5th and 97.5th percentiles obtained on the bootstrap samples. To summarize the improvements in Kaplan-Meier graft survival over time, we calculated the percent increase in 1-year and 5-year graft survival from the earliest era to the last era where follow-up was sufficient to ascertain survival at these time points. The 1-year percent improvements were calculated between eras 1995–1999 and 2014–2017. The 5-year percent improvements were calculated between eras 1995–1999 and 2010–2013 because 2010–2013 was the last era with observed 5-year follow-up. The percent improvement over era was calculated by type of donor, race, age group, and diagnosis of ESKD.

We then fitted multivariable Cox regression models to evaluate the associations between era and graft failure while adjusting for the following variables: recipient age, gender, diagnosis of ESKD, race, body mass index (BMI) category, insurance type, dialysis vintage, re-transplant status, donor age, and donor type (deceased vs. living donors). To account for multiple observations per recipient we used the robust sandwich estimator. To evaluate whether graft survival improvements over time differed across groups, we fitted models with two-way interactions between era and each of the following: deceased vs. living kidney donor, race, age group, and diagnosis group. The resulting hazard ratios were graphed to contrast the time effect across groups.

We also evaluated a Cox model of time to graft failure for deceased donor recipients only while adjusting for the variables mentioned above plus Kidney Donor Risk Index (KDRI), and a Cox model for live donor recipients only while adjusting for the variables mentioned above plus donor BMI, donor biological relationship to recipient, and donor history of hypertension. We also evaluated competing risks regression models of time to graft failure with death as a competing risk, and of time to death with graft failure as a competing risk. The models were adjusted for the same variables as in the time to graft failure (graft or death) mentioned above. We also evaluated Cox models of time to graft failure censored at death and time to death censored at graft failure.

To evaluate whether the graft survival improvements observed over time were limited to gains in 1-year graft survival, we fitted the Cox model of graft survival mentioned above only among patients whose graft survived to 1 year. In addition, we also fitted a piecewise exponential model with time of follow-up split into two intervals: year 1 posttransplant, and all the remaining survival follow-up. We tested the interactions between era and time interval within these models.

All data were analyzed with Linux SAS, version 9.4 (SAS Institute, Cary, NC). Graphics were produced with R version 3.5.1 (The R Foundation for Statistical Computing, Vienna, Austria).

3 | RESULTS

We included 331,216 kidney transplant recipients in the study, of which 210,665 (63.6%) were recipients of deceased donor kidneys, and 120,551 (36.4%) of living donors. Table 1 shows the recipient characteristics across the different eras. Transplant recipients in more recent eras were more likely to be older, of Black race, of Hispanic ethnicity, obese, diabetic, have longer dialysis vintage, and have Medicare coverage. During a median follow-up of 5.4 years, ranging from 0 to 25 years, there were 134,942 graft failures. Of these, 73,796 were graft losses and 61,146 were deaths. Table 2 and Figure 1 show Kaplan-Meier graft survival by era for deceased and living kidney donor transplant recipients, respectively. Note that the era 2010–2013 is the last era for which all recipients had the opportunity for complete a 5-year follow-up and therefore we highlight improvements in 5-year survival for that era. For deceased donor kidney transplant recipients, 1-year graft survival increased from 87.7% to 94.3% (7.5% improvement from era 1995–1999 to 2014–2017), and 5-year graft survival increased from 65.9% to 76.3% (15.7% improvement from era 1995–1999 to 2010–2013). In the case of living kidney donor transplant recipients, 1-year graft survival improved from 93.9% to 97.8% (4.2% improvement from era 1995–1999 to 2014–2017) and 5-year graft survival increased from 79.0% to 86.5% (9.5% improvement from era 1995–1999 to 2010–2013).

3.1 | Relative increase of graft survival over time by subgroups

Figure 2 and Table S1 show the percent or relative improvement in graft survival at 1 and 5 years posttransplant overall, and for different groups of interest. In the era 2010–2013, which is the last era for which all recipients had the opportunity to complete 5 years of follow-up, relative improvements were observed in 5-year survival compared to 1-year survival for all eras and subgroups. Of note, Black recipients had 25.5% improvement in 5-year graft survival and recipients age 65 years and older had a 20.9% improvement which is higher than the respective 1-year survival. Recipients of deceased and living kidney donors experienced a 15.7% and 9.5% improvement at 5 years, respectively. Diabetic patients demonstrated a 12.4% improvement, and those with a diagnosis of hypertension a 19.8% improvement. These 5-year rates were much higher than the relative improvement observed for the first-year posttransplant, indicating that changes in management and care in the current era may have had a greater impact on longer-term outcomes than within the first year posttransplant. Further, Table S2 shows the hazard ratios of graft failure from models adjusted for recipient age, gender, race, diagnosis, BMI, insurance type, dialysis time, re-transplant status, donor age and donor type (deceased vs. living donor). Compared to era 1995–1999, the overall hazard ratio for era 2000–2004 was 0.89 (95% CI: 0.88, 0.91), and continued to decrease over time, to 0.46 (95% CI: 0.45, 0.47) for the era 2014–2017. When we evaluated the same model only among those with 1-year graft survival, we found similar but slightly smaller effects, so that recipients in 2014–2017 had a hazard ratio of 0.52 (95% CI: 0.51, 0.54) compared to 1995–1999. We also found improvements in later eras when we evaluated graft failure censored at death and graft failure with death as a competing risk (Table 3). Similarly, improvements were also found when we ran separate models among living and deceased donor recipients while adjusting for additional variables (Table S3). Figure 3 shows adjusted hazard ratios of interactions between several recipient characteristics and transplant era on graft failure. The hazard ratios for each group in Figure 3 were calculated comparing 2014–2017 to the reference era 1995–1999. As shown in the graph, all groups, including different donor types, race, age, and diagnoses had significantly lower hazard ratios of graft failure in 2014–2017. When we evaluated the interactions between each of the variables in Figure 3 by continuous year rather than by era, we also found that all interactions were statistically significant (data not shown).

3.2 | Improvement of half-lives over time

Table 2 shows the observed and predicted half-life for deceased and living kidney donor transplant recipients. Observed median graft survival for deceased donor kidney transplants increased from 8.2 years in era 1995–1999 to 9.7 years in era 2005–2009, an 18.3% improvement over a 10-year period. The predicted half-life for those transplanted in era 2014–2017 is 11.7 years, representing a 43% increase in half-life over a 20-year period.

Likewise, for recipients of living donor kidney transplants, the observed actual median graft survival increased from 12.1 years to 12.9 years for those transplanted from 1995–1999 to 2000–2004, a 6.6% increase over a 5-year period. For those transplanted after 2009, median half-life has not been reached, but the projected half-life for those transplanted in era 2014–2017 is 19.2 years, representing a 59% improvement over a 20-year period.

3.3 | Exponential model evaluating the effect of first year vs. later follow-up

In the piecewise exponential model, we found main effects confirming that the first-year interval had increased rates of graft failure (estimate −0.28 indicating shorter time to failure, $p < .001$) when compared to the later interval (follow-up time after 1 year), and all later eras had reduced rates of graft failure (parameter estimates = 0.12,

TABLE 1 Recipient and donor characteristics by era of transplantation

Factor	Overall (*N* = 331,216)	1995–1999 (*N* = 55,287)	2000–2004 (*N* = 68,910)	2005–2009 (*N* = 77,164)	2010–2013 (*N* = 62,060)	2014–2017 (*N* = 67,795)
Donor type						
Deceased	210,665 (63.6)	37,103 (67.1)	40,130 (58.2)	47,453 (61.5)	39,815 (64.2)	46,164 (68.1)
Living	120,551 (36.4)	18,184 (32.9)	28,780 (41.8)	29,711 (38.5)	22,245 (35.8)	21,631 (31.9)
Previous kidney transplant	40,832 (12.3)	6,555 (11.9)	8,574 (12.4)	9,250 (12.0)	7,629 (12.3)	8,824 (13.0)
Recipient age (years)						
18–34	54,904 (16.6)	12,781 (23.1)	12,936 (18.8)	11,620 (15.1)	8,271 (13.3)	9,296 (13.7)
34–54	145,715 (44.0)	27,86 6 (50.4)	32,820 (47.6)	33,109 (42.9)	24,910 (40.1)	27,010 (39.8)
55–64	82,969 (25.0)	10,800 (19.5)	15,661 (22.7)	20,573 (26.7)	17,375 (28.0)	18,560 (27.4)
65–99	47,628 (14.4)	3,840 (6.9)	7,493 (10.9)	11,862 (15.4)	11,504 (18.5)	12,929 (19.1)
Recipient gender						
Female	131,384 (39.7)	22,254 (40.3)	27,950 (40.6)	30,271 (39.2)	24,300 (39.2)	26,609 (39.2)
Male	199,832 (60.3)	33,033 (59.7)	40,960 (59.4)	46,893 (60.8)	37,760 (60.8)	41,186 (60.8)
Recipient race						
White	179,347 (54.1)	33,751 (61.0)	40,113 (58.2)	42,025 (54.5)	31,961 (51.5)	31,497 (46.5)
Black	83,454 (25.2)	12,801 (23.2)	16,363 (23.7)	19,410 (25.2)	16,198 (26.1)	18,682 (27.6)
Hispanic	45,691 (13.8)	5,856 (10.6)	8,347 (12.1)	10,518 (13.6)	9,270 (14.9)	11,700 (17.3)
Other	22,724 (6.9)	2,879 (5.2)	4,087 (5.9)	5,211 (6.8)	4,631 (7.5)	5,916 (8.7)
Recipient BMI (kg/m^2)						
Missing	27,428 (8.3)	16,901 (30.6)	5,925 (8.6)	3,490 (4.5)	921 (1.5)	191 (0.28)
13–20	20,401 (6.2)	3,826 (6.9)	4,911 (7.1)	4,680 (6.1)	3,366 (5.4)	3,618 (5.3)
>20–25	90,309 (27.3)	14,505 (26.2)	20,945 (30.4)	21,240 (27.5)	15,808 (25.5)	17,811 (26.3)
>25–30	101,206 (30.6)	12,310 (22.3)	20,928 (30.4)	24,918 (32.3)	20,390 (32.9)	22,660 (33.4)
>30–35	62,335 (18.8)	5,510 (10.0)	11,210 (16.3)	15,439 (20.0)	14,328 (23.1)	15,848 (23.4)
>35–50	29,537 (8.9)	2,235 (4.0)	4,991 (7.2)	7,397 (9.6)	7,247 (11.7)	7,667 (11.3)
Recipient insurance						
VA and other	6,762 (2.0)	1,788 (3.2)	1,306 (1.9)	1,329 (1.7)	1,040 (1.7)	1,299 (1.9)
Private	124,048 (37.5)	20,438 (37.0)	29,300 (42.5)	30,760 (39.9)	22,331 (36.0)	21,219 (31.3)
Medicaid	15,046 (4.5)	2,543 (4.6)	3,033 (4.4)	3,521 (4.6)	2,324 (3.7)	3,625 (5.3)
Medicare	185,360 (56.0)	30,518 (55.2)	35,271 (51.2)	41,554 (53.9)	36,365 (58.6)	41,652 (61.4)
Diagnosis group						
Diabetes	81,829 (24.7)	12,143 (22.0)	15,835 (23.0)	19,101 (24.8)	16,608 (26.8)	18,142 (26.8)
Hypertension	66,103 (20.0)	8,325 (15.1)	11,840 (17.2)	17,223 (22.3)	13,782 (22.2)	14,933 (22.0)
Dialysis time						
Preemptive	58,214 (17.6)	8,250 (14.9)	13,180 (19.1)	14,389 (18.6)	11,082 (17.9)	11,313 (16.7)
<=12 months	55,751 (16.8)	12,554 (22.7)	12,619 (18.3)	12,158 (15.8)	9,248 (14.9)	9,172 (13.5)
>12–24 months	52,702 (15.9)	11,456 (20.7)	12,084 (17.5)	12,342 (16.0)	8,484 (13.7)	8,336 (12.3)
>24–48 months	75,507 (22.8)	13,358 (24.2)	16,101 (23.4)	18,237 (23.6)	14,307 (23.1)	13,504 (19.9)
>48–72 months	44,736 (13.5)	5,268 (9.5)	8,373 (12.2)	10,470 (13.6)	9,574 (15.4)	11,051 (16.3)
>72 months	44,306 (13.4)	4,401 (8.0)	6,553 (9.5)	9,568 (12.4)	9,365 (15.1)	14,419 (21.3)
Donor age[a]	39.0 ± 15.1	36.8 ± 15.7	38.4 ± 14.9	39.8 ± 14.9	39.9 ± 14.9	39.7 ± 14.9

Statistics presented as mean ±SD or N (column %). All variables significantly different across time eras ($P < .01$) using linear regression, Cochran-Armitage Trend Test, and Cochran-Mantel-Haenszel Statistics.

[a]Missing donor age for seven transplants.

0.36, 0.66, and 0.96 indicating longer time to failure for 2000–2004, 2005–2009, 2010–2013, and 2014–2017 vs. 1995–1999 respectively, all $p < .001$). We also found a significant interaction between era and time interval ($p < .001$), suggesting that when compared to 1995–1999 all later eras have differentially reduced rates of graft failure during year 1 of follow-up (interaction effect estimate for

TABLE 2 Kaplan-Meier graft survival by era of transplantation and half-lives

Donor type	N	1 year	3 years	5 years	10 years	Median survival in years
Deceased						
1995–1999	37103	87.7 (87.4, 88.1)	77.4 (76.9, 77.8)	65.9 (65.4, 66.4)	42.6 (42.0, 43.1)	8.2 (8.1, 8.3)
2000–2004	40130	88.9 (88.6, 89.2)	78.3 (77.9, 78.7)	67.8 (67.3, 68.3)	44.9 (44.4, 45.4)	8.8 (8.6, 8.9)
2005–2009	47453	91.0 (90.7, 91.2)	82.0 (81.7, 82.4)	72.3 (71.8, 72.7)	48.8 (48.3, 49.3)	9.7 (9.6, 9.8)
2010–2013	39815	93.0 (92.8, 93.3)	85.3 (85.0, 85.7)	76.3 (75.9, 76.7)	51.6 (50.0, 53.3)	10.5 (10.4, 10.7)[a]
2014–2017	46164	94.3 (94.1, 94.5)	87.1 (86.8, 87.4)	78.1 (77.5, 78.6)		11.7 (11.4, 12.1)[a]
Live						
1995–1999	18184	93.9 (93.6, 94.3)	87.2 (86.7, 87.7)	79.0 (78.4, 79.6)	57.9 (57.1, 58.7)	12.1 (11.9, 12.3)
2000–2004	28780	94.8 (94.5, 95.1)	88.1 (87.8, 88.5)	80.6 (80.1, 81.1)	60.7 (60.1, 61.3)	12.9 (12.7, 13.1)
2005–2009	29711	96.1 (95.9, 96.3)	90.7 (90.3, 91.0)	84.1 (83.7, 84.5)	65.0 (64.5, 65.6)	13.9 (13.7, 14.2)
2010–2013	22245	97.1 (96.9, 97.3)	92.5 (92.1, 92.8)	86.5 (86.1, 87.0)	67.6 (65.6, 69.6)	15.7 (15.2, 16.1)[a]
2014–2017	21631	97.8 (97.6, 98.0)	93.9 (93.5, 94.2)	88.0 (87.3, 88.7)		19.2 (18.1, 20.7)[a]

[a]Predicted median survival derived from linear regression analysis for transplant recipients where median survival has not been observed at latest follow-up (March 2020), and predicted 95% confidence intervals obtained from 1,000 bootstrap sample analysis.

interval 1 were 0.09, 0.16, 0.17, and 0.10 for 2000–2004, 2005–2009, 2010–2013, and 2014–2017 vs. 1995–1999 respectively, all $p \leq .001$). Results confirmed a statistically significant improvement in survival over the eras in the first year posttransplant as well as beyond the first year with larger estimated improvements in long-term outcomes. The overall predicted 10-year graft survival according to the exponential piecewise model were 42.2% in 1995–1999, 47.7% in 2000–2004, 54.1% in 2005–2009, 61.5% in 2010–13, and 67.8% in 2014–2017.

4 | DISCUSSION

The results of this study are notable in demonstrating that long-term kidney transplant outcomes are steadily improving over time. In other words, significant prolongation of graft survival is now appreciable well beyond the first year following transplantation. This observed improvement is due to both better graft and patient survival. Kidney transplant recipients of deceased and living donor kidneys experience one-year graft survival above 94% and 97%, respectively. More importantly, observed graft survival half-lives from 2005 to 2009 were greater than 9 years and 13 years, respectively. Furthermore, projected half-lives for deceased donor and living kidney donor transplant recipients are now expected to be above 11 years and 19 years, respectively. Finally, this improvement in graft survival is more pronounced in minority race/ethnic groups compared to Caucasians, diabetic and hypertensive recipients compared to patients with other causes of kidney disease, and older compared to younger transplant recipients. Thus the difference in survival among these relatively higher-risk transplant groups is narrowing with respect to long-term outcomes, and should translate into a greater relative benefit of transplantation for the treatment of patients with ESKD. This information is relevant and timely as it provides further evidence supporting the need to pursue kidney transplantation among all eligible patient groups as stated by the Executive Order of 2019.[1,2]

Regulatory entities like UNOS measure kidney transplant success as 1-year and 3-year graft survival. Public and private health care insurances and payers also utilize these metrics to determine participation of their insured patients in transplant programs. These metrics, especially 1-year graft and patient survival, are limited to short-term events, hence utilized by regulatory and healthcare payers as a marker of recent transplant center outcomes quality. However, among the many inherent limitations of using 1-year graft and patient survival, is that success rates are now nearly maximized, and therefore a small percentage change in outcomes may not be clinically impactful but may affect a program's status.[14,15] Particularly from the patient perspective, the more important endpoint is the success of longer-term outcomes, both patient and allograft survival.[16] This is even more relevant when considering the high mortality rate of patients who remain on the waiting list versus those receiving a kidney transplant. There is also a high rate of mortality following kidney transplant failure with return to dialysis, relative to incident dialysis patients without transplantation,[17] and conversely, there is a patient survival benefit that correlates with a functioning allograft over time. Thus extending survival outcomes to a greater number of years after transplantation is very important for the healthcare system, transplantation community and prospective end-organ kidney disease patients seeking a kidney transplant.

It has been repeatedly claimed that short-term outcomes in kidney transplantation do not translate into longer-term outcomes. At the beginning of the century, work by Meier-Kriesche et al showed that while 1-year graft survival was improving in the US during the 1990 s compared to the immediate postcyclosporine era, but long-term graft outcomes were significantly lagging behind.[10,11] A

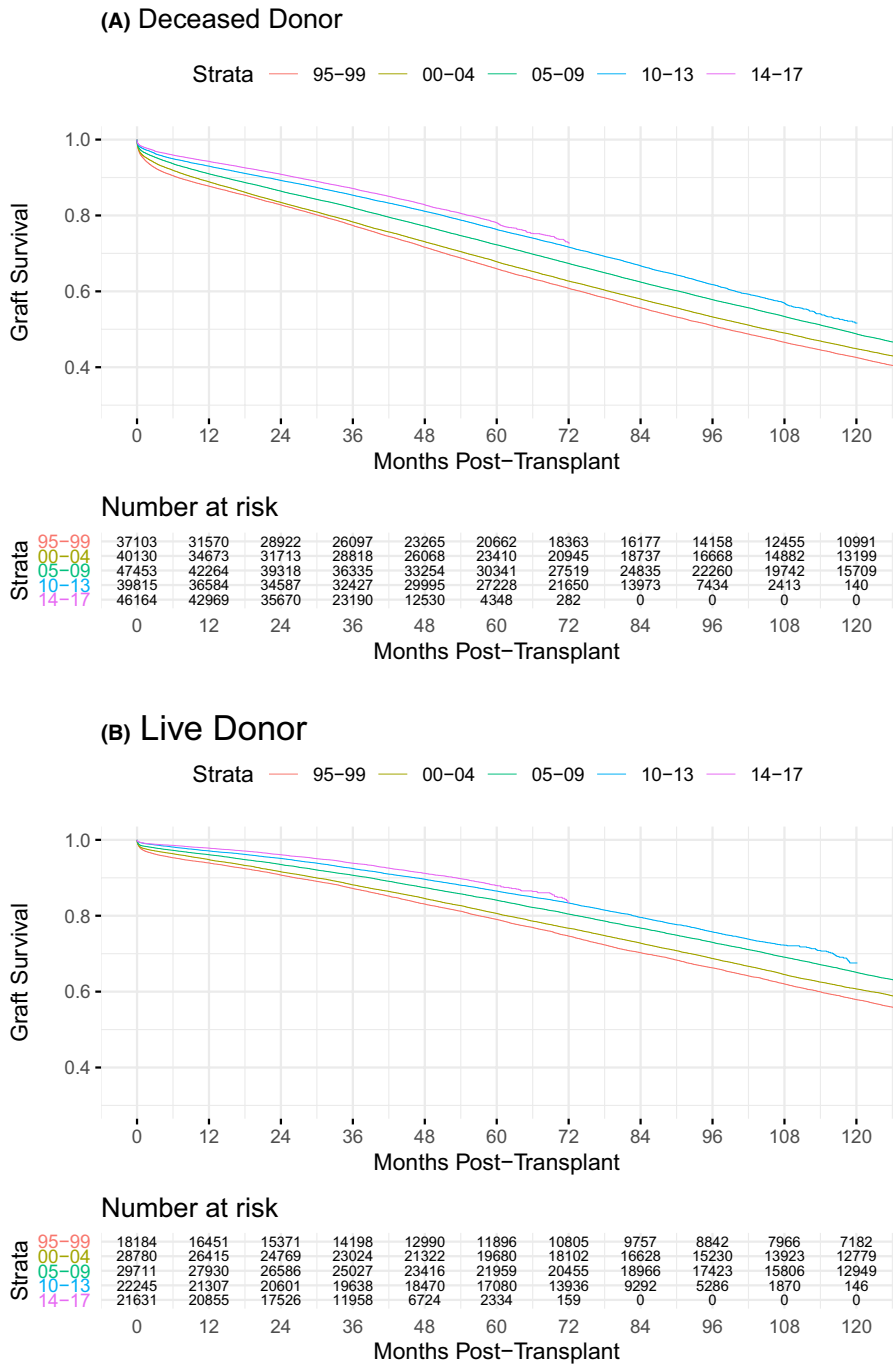


FIGURE 1 Kaplan-Meier curves for graft failure for recipients of deceased kidney donors (A) and of living kidney donors (B)

reappraisal of this work was then published by Lamb et al in 2011, who again showed excellent short-term results, and began to note some modest improvements in longer-term outcomes.[12] In our study, using the same registry but now with a larger cohort of transplanted recipients and a longer and more contemporary time frame (from 1995 to 2017) and follow-up period (as of March 2020), we demonstrate that there has been significant continuous improvement in long-term outcomes that are highly clinically impactful. This improvement is due to increase not only in graft survival but also in patient survival, suggesting better kidney transplant care and overall patient care. This information is enlightening and encouraging for the transplant community at large, and is reassuring that continual progress is being made.

Another way of depicting long-term success is by using half-life of the graft, which facilitates interpretation by prospective transplant recipients regarding expected transplant survival. In this study we showed that actual half-lives for both deceased and living kidney donor transplant recipients continue to improve. Actual half-lives for deceased donors transplanted between 2005 and 2009 (last entire cohort with accrued outcomes) were 9.7 years versus 8.2 years for those transplanted from 1995 to 1999 (18% improvement in a 10-year period). In recipients of living kidney donors, actual half-lives

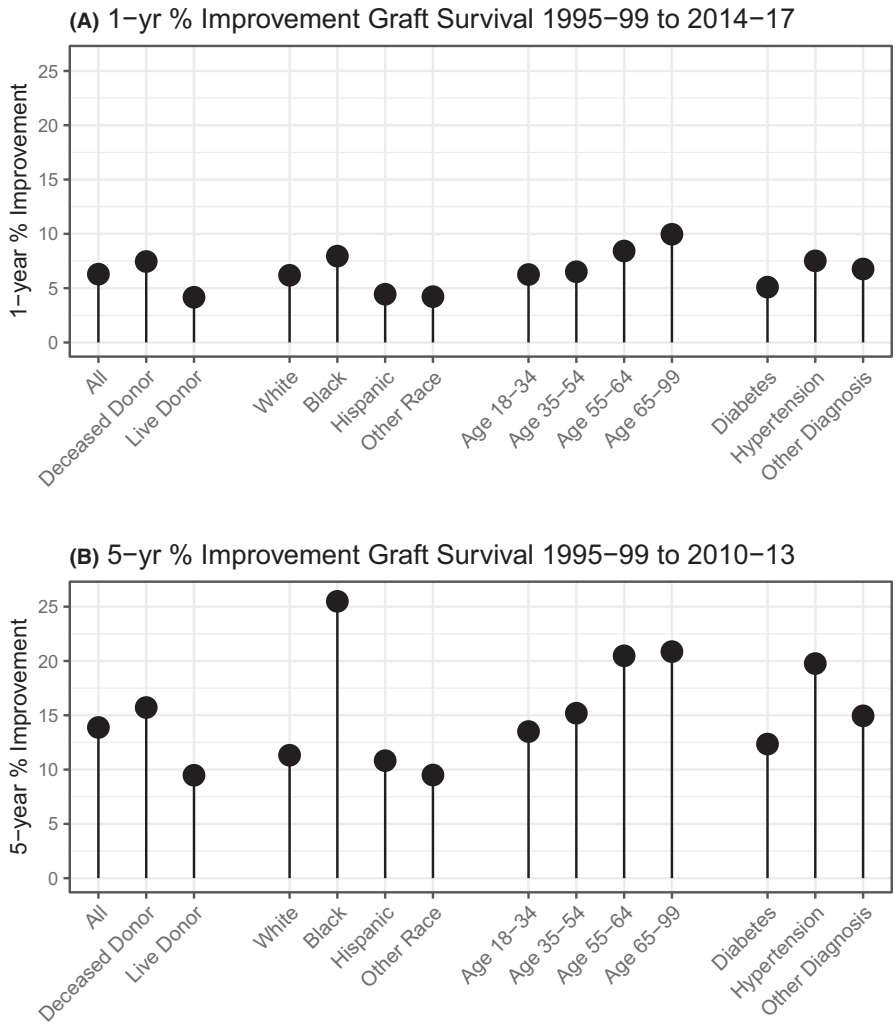


FIGURE 2 Graphs depicting the relative improvement in graft survival for overall and various subpopulations at 1 year from eras 1995–1999 to 2014–2017 (A) and 5 years from eras 1995–1999 to 2010–2013 (B)

TABLE 3 Adjusted Cox proportional hazards model of time to graft failure censored at death, competing risks analysis of graft failure with death as a competing risk, Cox proportional hazards model of time to death censored at graft failure, and competing risks analysis of death with graft failure as a competing risk

	Graft failure censored at death HR (95% CI)	Graft failure with death as a competing risk SHR (95% CI)	Mortality censored at graft failure HR (95% CI)	Mortality with graft failure as a competing risk SHR (95% CI)
Era (year)				
1995–1999	Ref	Ref	Ref	Ref
2000–2004	0.86 (0.84, 0.88)	0.85 (0.83, 0.86)	0.92 (0.90, 0.94)	0.91 (0.89, 0.93)
2005–2009	0.69 (0.67, 0.71)	0.67 (0.66, 0.69)	0.72 (0.71, 0.74)	0.71 (0.70, 0.73)
2010–2013	0.54 (0.53, 0.56)	0.52 (0.51, 0.54)	0.59 (0.58, 0.61)	0.56 (0.54, 0.57)
2014–2017	0.43 (0.41, 0.44)	0.40 (0.39, 0.41)	0.51 (0.49, 0.53)	0.46 (0.44, 0.48)

Models adjusted for recipient age, gender, diagnosis of ESKD, race, body mass index (BMI) category, insurance type, dialysis vintage, re-transplant status, donor age, and donor type (deceased vs. living donors).

were 13.9 years versus 12.1 years for those transplanted from 2005 to 2009 compared to 1995 to 1999 (15% improvement over a 5-year period). This study has the longest follow-up of all published data and while it includes many transplant recipients with currently functioning grafts, the actual half-lives show continuous improvement over time. For recipients of transplants of deceased donors occurring after 2009, and after 2005 for recipients of organs from living kidney donors, we calculated projected half-lives. While this metric may have inherent

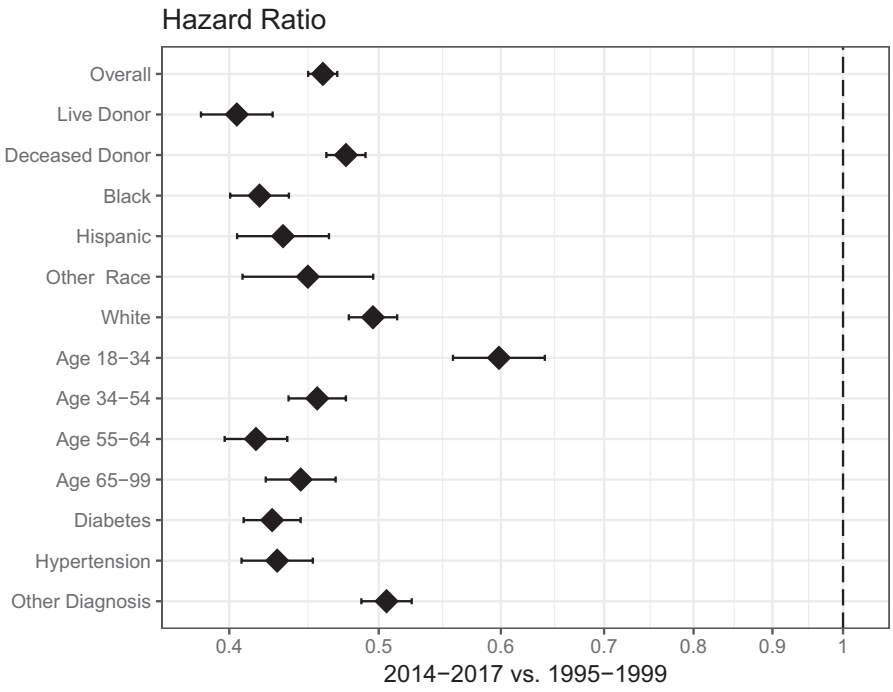


FIGURE 3 Graph depicting hazard ratios of graft failure for overall and various subpopulations from eras 1995-1999 to 2014-2017

uncertainties, the information is still very useful for evaluating progress. We present projected deceased donor kidney graft survival of 11.7 years for those transplanted from 2014 to 2017, which represents a 43% improvement compared to those transplanted from 1995 to 1999. In living kidney donors, projected graft survival was 19.2 years for those transplanted from 2014 to 2017, a 59% improvement compared to the 1995–1999 era. When compared to projections made in prior studies,[9,10] this study has much longer actual follow-up than the studies by Meier-Kriesche et al and Lamb et al, making the projected half-lives of this report more stable and reliable. It is also important to note that projections might also vary to some degree depending on the statistical approach used, therefore, it is important to remark that this study utilized several models to ascertain that reported projections are very similar independent of the model used. Regardless of the assumptions required for calculating projected half-lives, this study shows that there has been marked significant progress in augmenting long-term graft survival over time.

Another important observation is that not all subgroups have improved long-term survival to the same degree. We show more pronounced improvement in long-term outcomes in older rather than younger people. Advances in the care of comorbidities that disproportionally afflict older populations such as diabetes mellitus, cardiovascular disease, and malignancies likely benefit older individuals to a larger degree than younger ones. In fact, diabetic and hypertensive transplant recipients have experienced better survival gains than those with other diagnosis of kidney disease. While Black recipients do not fair equally well as non-Blacks,[18-22] our study shows that graft survival has been improving to a much larger degree in this population compared to the non-Black population, narrowing the gap that has always existed in kidney transplantation. Purnell et al. previously described a narrowing in the gap in outcomes between Blacks and white kidney recipients, analyzing registry data from 1990 to 2012.[21] Declines in survival differences in deceased donor transplant appeared to be driven in part by a relative reduction in early rejection and delayed graft function over time in Blacks. Still, disparities remain, particularly with access to living donation, which has actually declined in Black recipients relative to whites in the recent era.[23]

Finally, grafts from deceased donors are also demonstrating a greater improvement in graft survival compared to those of living donors. This is likely indicative of improvements in organ procurement and preservation, better management of delayed graft function and other transplant complications, and advances in the care of comorbidities affecting transplant recipients with a longer dialysis vintage.[24,25] This is in spite of an increase in the utilization of higher-risk organs.[6,26-29] In summary, the progress in graft survival is not homogenous but appears to be preferentially benefiting those populations traditionally considered disadvantaged or at higher risk, thus narrowing the gap in terms of disparities seen in transplant outcomes.

In conclusion, kidney transplant survival has improved dramatically over the past three decades in the US. In past eras, the improvement in graft survival was limited to outcomes within the first year of transplantation. It is now evident that long-term graft survival is steadily improving. More importantly, traditionally disadvantaged populations like minorities, older patients, and diabetics and hypertensive patients are making more progress with regards to prolongation of viable kidney transplants. This comes at a critical time when the Presidential Executive Order of "Advancing American Kidney Health" is to be implemented to increase transplant rates and when the number of kidney transplants is already at an all-time high.

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DISCLOSURE

The authors of this manuscript have no conflicts of interest to disclose as described by the *American Journal of Transplantation*.

DATA AVAILABILITY STATEMENT

The data were obtained from the public SRTR registry.

ORCID

Emilio D. Poggio https://orcid.org/0000-0003-1492-5103
Jesse D. Schold https://orcid.org/0000-0002-5341-7286

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SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section.

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